

TRAINED AGENTIC CONTEXT MANAGEMENT

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ABSTRACT

We study long context language models. Instead of training long context natively, or designing a long context harness, we train a model over the simplest possible harness: a tool to call itself with any specified prompt and a tool to read tokens in a range from the input context. We finetune Qwen3.6-35B-A3B over this harness. With only 8,000 tokens of context, our small model outperforms GPT-5.4 with 1M tokens of context on the OOLONG-synth benchmark when document length exceeds 40K tokens.

1 INTRODUCTION

As language models boast impressive breakthroughs (Luong & Lockhart, 2025; Wei, 2025; OpenAI, 2025; Lin & Cheng, 2025; OpenAI, 2026c;b), increasing focus is placed on their limitations. The most salient resource constraint of a language model is its context length. Models can process and produce tokens within this limit, but the limit is hard. Model producers have identified this constraint and placed great emphasis on extending it as large as possible, with some model builders touting 10M (Dangel, 2026; Meta AI, 2025; Zhu et al., 2026) and even 100M (Magic, 2024) token context lengths (for reference, the length of the first Harry Potter book is about 100,000 tokens, the full U.S. tax code roughly 3M, and the Linux kernel source code about 300M).

While native context scaling is appealing as a concrete engineering problem, it is not the only approach to producing a model capable of processing long documents. In 2021, OpenAI fine-tuned GPT-3 to summarize sections of books, and then summarize those summaries, achieving competitive performance with human book summaries on a model with a context length of just 2,048 tokens (Wu et al., 2021). As the field has matured, agentic harnesses have become viable. MemGPT treats context similarly to an operating system, relying on LLM-invoked function calls to push relevant information in and out of the context window to external memory, effectively achieving expanded virtual context (Packer et al., 2023). More recently, Recursive Language Models introduced a harness that utilizes LLM’s coding prior to call itself in a REPL environment, achieving stronger performance than base models on retrieval and aggregation tasks both within context limits, but also beyond it, expanding effective context by $\sim 4\times$ (Zhang et al., 2025).

While handcrafted prompts and programmatic environments can be effective in solving long context problems without native long context training, it begs an obvious question: if models can, for example, solve all the problems at the ICPC World Finals (OpenAI, 2025), or more generally, implement requested features in codebases well beyond context limits (Marten et al., 2026; OpenAI, 2026a; Anthropic, 2026), can they not be *trained* to manage context effectively, *without human-authored harness crutches*? Such model training internalizes decisions external harnesses hardcode, offering the potential to scale much more generally.

We answer this question in this paper, affirmatively. We consider the simplest possible agentic context harness, defined below.

1.1 HARNESS

Our harness has two tools:

1. `read_chunk(start, end)`: Read a slice of the document and return the decoded text of tokens `[start, end)`.

2. `spawn_subagent(subtask)`: Spawn a fresh-context copy of yourself to solve `subtask`; returns the child's `\boxed{}` answer.

Unmodified long-context prompts are provided to the model over a document accessed via the above harness. The system prompt is propagated to child agents and includes the above harness definition as well as the original prompt¹; child agents' user prompt is specified by the calling model.

1.2 RESULTS

We finetune Qwen3.6-35B-A3B on the above harness and achieve the following results:

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2. `spawn_subagent(subtask)`: Spawn a fresh-context copy of yourself to solve `subtask`; returns the child's `\boxed{}` answer.

We train a model on this harness to answer unmodified long-context prompts over a document accessed via the above harness.

We propagate the system prompt to subagents, which

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The formatting instructions contained in these style files are summarized in sections 3, 4, and 5 below.

¹The full system

3 GENERAL FORMATTING INSTRUCTIONS

The text must be confined within a rectangle 5.5 inches (33 picas) wide and 9 inches (54 picas) long. The left margin is 1.5 inch (9 picas). Use 10 point type with a vertical spacing of 11 points. Times New Roman is the preferred typeface throughout. Paragraphs are separated by 1/2 line space, with no indentation.

Paper title is 17 point, in small caps and left-aligned. All pages should start at 1 inch (6 picas) from the top of the page.

Authors' names are set in boldface, and each name is placed above its corresponding address. The lead author's name is to be listed first, and the co-authors' names are set to follow. Authors sharing the same address can be on the same line.

Please pay special attention to the instructions in section 5 regarding figures, tables, acknowledgments, and references.

There will be a strict upper limit of **9 pages** for the main text of the initial submission, with unlimited additional pages for citations. This limit will be expanded to **10 pages** for rebuttal/camera ready.

4 HEADINGS: FIRST LEVEL

First level headings are in small caps, flush left and in point size 12. One line space before the first level heading and 1/2 line space after the first level heading.

4.1 HEADINGS: SECOND LEVEL

Second level headings are in small caps, flush left and in point size 10. One line space before the second level heading and 1/2 line space after the second level heading.

4.1.1 HEADINGS: THIRD LEVEL

Third level headings are in small caps, flush left and in point size 10. One line space before the third level heading and 1/2 line space after the third level heading.

5 CITATIONS, FIGURES, TABLES, REFERENCES

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5.1 CITATIONS WITHIN THE TEXT

Citations within the text should be based on the `natbib` package and include the authors' last names and year (with the "et al." construct for more than two authors). When the authors or the publication are included in the sentence, the citation should not be in parenthesis using `\citet{}` (as in "See Hinton et al. (2006) for more information."). Otherwise, the citation should be in parenthesis using `\citep{}` (as in "Deep learning shows promise to make progress towards AI (Bengio & LeCun, 2007).").

The corresponding references are to be listed in alphabetical order of authors, in the REFERENCES section. As to the format of the references themselves, any style is acceptable as long as it is used consistently.

5.2 FOOTNOTES

Indicate footnotes with a number² in the text. Place the footnotes at the bottom of the page on which they appear. Precede the footnote with a horizontal rule of 2 inches (12 picas).³

²Sample of the first footnote

³Sample of the second footnote

Table 1: Sample table title

PART	DESCRIPTION
Dendrite	Input terminal
Axon	Output terminal
Soma	Cell body (contains cell nucleus)

5.3 FIGURES

All artwork must be neat, clean, and legible. Lines should be dark enough for purposes of reproduction; art work should not be hand-drawn. The figure number and caption always appear after the figure. Place one line space before the figure caption, and one line space after the figure. The figure caption is lower case (except for first word and proper nouns); figures are numbered consecutively.

Make sure the figure caption does not get separated from the figure. Leave sufficient space to avoid splitting the figure and figure caption.

You may use color figures. However, it is best for the figure captions and the paper body to make sense if the paper is printed either in black/white or in color.

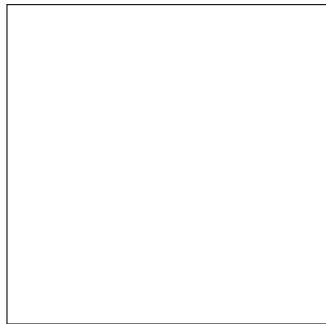


Figure 1: Sample figure caption.

5.4 TABLES

All tables must be centered, neat, clean and legible. Do not use hand-drawn tables. The table number and title always appear before the table. See Table 1.

Place one line space before the table title, one line space after the table title, and one line space after the table. The table title must be lower case (except for first word and proper nouns); tables are numbered consecutively.

6 DEFAULT NOTATION

In an attempt to encourage standardized notation, we have included the notation file from the textbook, *Deep Learning* Goodfellow et al. (2016) available at https://github.com/goodfeli/dlbook_notation/. Use of this style is not required and can be disabled by commenting out `math_commands.tex`.

Numbers and Arrays

216	$\mathbf{a}$	A scalar (integer or real)
217	$\mathbf{a}$	A vector
218	$\mathbf{A}$	A matrix
219	$\mathbf{A}$	A matrix
220	$\mathbf{A}$	A tensor
221	$\mathbf{I}_n$	Identity matrix with n rows and n columns
222	$\mathbf{I}$	Identity matrix with dimensionality implied by context
223	$\mathbf{e}^{(i)}$	Standard basis vector $[0, \dots, 0, 1, 0, \dots, 0]$ with a 1 at position i
224	$\text{diag}(\mathbf{a})$	A square, diagonal matrix with diagonal entries given by $\mathbf{a}$
225	$\mathbf{a}$	A scalar random variable
226	$\mathbf{a}$	A vector-valued random variable
227	$\mathbf{A}$	A matrix-valued random variable
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Sets and Graphs

233	$\mathbb{A}$	A set
234	$\mathbb{R}$	The set of real numbers
235	$\{0, 1\}$	The set containing 0 and 1
236	$\{0, 1, \dots, n\}$	The set of all integers between 0 and n
237	$[a, b]$	The real interval including a and b
238	$(a, b]$	The real interval excluding a but including b
239	$\mathbb{A} \setminus \mathbb{B}$	Set subtraction, i.e., the set containing the elements of $\mathbb{A}$ that are not in $\mathbb{B}$
240	$\mathcal{G}$	A graph
241	$\text{Pa}_{\mathcal{G}}(\mathbf{x}_i)$	The parents of $\mathbf{x}_i$ in $\mathcal{G}$
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Indexing

248	a_i	Element i of vector $\mathbf{a}$, with indexing starting at 1
249	$\mathbf{a}_{-i}$	All elements of vector $\mathbf{a}$ except for element i
250	$A_{i,j}$	Element i, j of matrix $\mathbf{A}$
251	$\mathbf{A}_{i,:}$	Row i of matrix $\mathbf{A}$
252	$\mathbf{A}_{:,i}$	Column i of matrix $\mathbf{A}$
253	$\mathbf{A}_{i,j,k}$	Element (i, j, k) of a 3-D tensor $\mathbf{A}$
254	$\mathbf{A}_{:,:,i}$	2-D slice of a 3-D tensor
255	$\mathbf{a}_i$	Element i of the random vector $\mathbf{a}$
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Calculus

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270	$\frac{dy}{dx}$	Derivative of y with respect to x
271	$\frac{\partial y}{\partial x}$	Partial derivative of y with respect to x
272	$\nabla_{\mathbf{x}} y$	Gradient of y with respect to $\mathbf{x}$
273	$\nabla_{\mathbf{X}} y$	Matrix derivatives of y with respect to $\mathbf{X}$
274	$\nabla_{\mathbf{x}} y$	Tensor containing derivatives of y with respect to $\mathbf{X}$
275	$\frac{\partial f}{\partial \mathbf{x}}$	Jacobian matrix $\mathbf{J} \in \mathbb{R}^{m \times n}$ of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$
276	$\nabla_{\mathbf{x}}^2 f(\mathbf{x})$ or $\mathbf{H}(f)(\mathbf{x})$	The Hessian matrix of f at input point $\mathbf{x}$
277	$\int f(\mathbf{x}) d\mathbf{x}$	Definite integral over the entire domain of $\mathbf{x}$
278	$\int_{\mathbb{S}} f(\mathbf{x}) d\mathbf{x}$	Definite integral with respect to $\mathbf{x}$ over the set $\mathbb{S}$
279	Probability and Information Theory	
280	$P(\mathbf{a})$	A probability distribution over a discrete variable
281	$p(\mathbf{a})$	A probability distribution over a continuous variable, or over a variable whose type has not been specified
282	$\mathbf{a} \sim P$	Random variable $\mathbf{a}$ has distribution P
283	$\mathbb{E}_{\mathbf{x} \sim P}[f(\mathbf{x})]$ or $\mathbb{E}f(\mathbf{x})$	Expectation of $f(\mathbf{x})$ with respect to $P(\mathbf{x})$
284	$\text{Var}(f(\mathbf{x}))$	Variance of $f(\mathbf{x})$ under $P(\mathbf{x})$
285	$\text{Cov}(f(\mathbf{x}), g(\mathbf{x}))$	Covariance of $f(\mathbf{x})$ and $g(\mathbf{x})$ under $P(\mathbf{x})$
286	$H(\mathbf{x})$	Shannon entropy of the random variable $\mathbf{x}$
287	$D_{\text{KL}}(P \ Q)$	Kullback-Leibler divergence of P and Q
288	$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian distribution over $\mathbf{x}$ with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$
289	Functions	
290	$f : \mathbb{A} \rightarrow \mathbb{B}$	The function f with domain $\mathbb{A}$ and range $\mathbb{B}$
291	$f \circ g$	Composition of the functions f and g
292	$f(\mathbf{x}; \boldsymbol{\theta})$	A function of $\mathbf{x}$ parametrized by $\boldsymbol{\theta}$. (Sometimes we write $f(\mathbf{x})$ and omit the argument $\boldsymbol{\theta}$ to lighten notation)
293	$\log x$	Natural logarithm of x
294	$\sigma(x)$	Logistic sigmoid, $\frac{1}{1 + \exp(-x)}$
295	$\zeta(x)$	Softplus, $\log(1 + \exp(x))$
296	$\ \mathbf{x}\ _p$	L^p norm of $\mathbf{x}$
297	$\ \mathbf{x}\ $	L^2 norm of $\mathbf{x}$
298	x^+	Positive part of x , i.e., $\max(0, x)$
299	$\mathbf{1}_{\text{condition}}$	is 1 if the condition is true, 0 otherwise

7 FINAL INSTRUCTIONS

Do not change any aspects of the formatting parameters in the style files. In particular, do not modify the width or length of the rectangle the text should fit into, and do not change font sizes (except perhaps in the REFERENCES section; see below). Please note that pages should be numbered.

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Otherwise, please generate your PostScript and PDF files with the following commands:

```
dvips mypaper.dvi -t letter -Ppdf -G0 -o mypaper.ps
ps2pdf mypaper.ps mypaper.pdf
```

8.1 MARGINS IN LATEX

Most of the margin problems come from figures positioned by hand using `\special` or other commands. We suggest using the command `\includegraphics` from the `graphicx` package. Always specify the figure width as a multiple of the line width as in the example below using `.eps` graphics

```
\usepackage[dvips]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.eps}
```

or

```
\usepackage[pdftex]{graphicx} ...
\includegraphics[width=0.8\linewidth]{myfile.pdf}
```

for `.pdf` graphics. See section 4.4 in the graphics bundle documentation (<http://www.ctan.org/tex-archive/macros/latex/required/graphics/grfguide.ps>)

A number of width problems arise when LaTeX cannot properly hyphenate a line. Please give LaTeX hyphenation hints using the `\-` command.

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In this work, we used generative AI tools for [tasks with required disclosure]. We have not used generative AI tools for [other tasks with required disclosure], and [the rest of the required disclosure tasks] are not applicable to this work. Additionally, we used generative AI tools for [tasks with recommended disclosure]. We have reviewed all AI-assisted work. [Elaborate. For example, “we checked LLM-generated research ideas for potential plagiarism through a manual literature survey”, “LLM-generated code was verified and tested for correctness by 2 authors”, etc.]. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

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AUTHOR CONTRIBUTIONS

If you'd like to, you may include a section for author contributions as is done in many journals. This is optional and at the discretion of the authors.

ACKNOWLEDGMENTS

Use unnumbered third level headings for the acknowledgments. All acknowledgments, including those to funding agencies, go at the end of the paper.

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A APPENDIX

You may include other additional sections here.